

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 322 — Past Questions & Materials

by Renance

Course: MAR 322

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — MAR 322 Marine Engrg. Design

Level 300 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF TECHNOLOGY

DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: Second Semeste: ce SO COURSE CODE: MAR 322

SESSION: 2019/2020 q es }

J COURSE TITLE: Marine Engrg. Design

DATE: 19/07/2021 DREEZA TIME ALLOWED: 2 Hrs

INSTRUCTION: ATTEMPT ANY 4 QUESTIONS

Question 1

- i. What is Ship Design? (2.5 marks)
- li. Describe the overall ship design processes. (10 marks)
- u. List any fifteen (15) Marine engineering design and analysis software program studied. (5 marks)

Question 2

- i. With the aid of a chart, describe the classification of offshore structures (7.5 marks)
- ii. Discuss the design Stages of offshore structures (10 marks)

Question 3

- 1s What is Marine Piling (2 marks)
- ii. Discuss the types of Pile Hammer studied (5 marks)
- iii. A pile of 1000 kg is driven into the seabed by the blows of a pile hammer of mass 2000 kg and falling on the head of the pile from a height of | m, in each blow. If the average resistance of the ground is 200 KN, Neglecting all fluid friction, calculates:
 - (a) The pile penetration per blow,
 - (b) The loss of energy at impact, and
 - (c) The Duration of time during which the pile is in motion (8.5 marks)

Question 4

- i. What is Bill of Engineering Measurement and Evaluation (BEME)? ae marks)
- ii. An estimate for the construction of a Cruise liner is given below. All cost is in US dollars (USD).
 - 1 Cost of the Hull is 20,000
 2. Cost of the super-structure is 60% more than the cost of the hull
 - 3 Four (4) power plants are needed, each cost 10% the cost of the hull
 4. | The Deck Machinery cost 20% the cost of the super-structure
 5. Ten (10) Navigational equipment, each cost 5% the cost of the hull

6. The Labour cost 40% more than the cost of the hull —

Develop a BEME for the project = K (é it

iii. | Describe the components of a Subsea System ;

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Question 5 of a container vessel is show below,

i i illage in the structure iS St

Se eee i external forces acting at these \

Determine the displacements at nodes; B,C and D if the \

yints are 10 KN, 25 KN and 15 KN respectively. ; — \

Take coefficient Of stiffness for the elements as K, = 2000 KN/m, K; 1500 KN/m.

K; = 1200 KN/m, K, = 1800 KN/m

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(10 marks)

ii. Explain the types of Steel Structural Failure studied. (7.5 marks)

Question 6

4, The Figure below is a section of an Offshore Frame Structure. Given that the external

loads, F3 and F4 are both 20 KN, The Area, A = 100 mm², Young Modulus, E = 200

KN/mmr', and for elements 'a' and 'b', Length, L = 2000 mm, Analyze the structure.

c Structure

Fyg or Fy

=> F ess

ia 8 orF.

(17.5 marks)

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